

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester II (NEP) Examination 2024-25

DEPARTMENT OF COMPUTER SCIENCE**Skill Enhancement Course (SEC)****Paper Code: CSCSE21 — Mobile Application Development**

Time: 2:30 Hours

Maximum Marks: 70

Paper Code Ref: CECNF/N/2024-25/21

*Instruction: Write your Roll No. at the top immediately on the receipt of this question paper.**Note: Question 1 is compulsory. Attempt any five from the remaining six questions.***Question 1** Choose the correct option from the following questions: [10 × 2 = 20]

- (a) What is Android?
- (i) A mobile device
 - (ii) An operating system
 - (iii) A web browser
 - (iv) A programming language
- (b) Which file is the entry point for an Android application?
- (i) `MainActivity.kt`
 - (ii) `AndroidManifest.xml`
 - (iii) `strings.xml`
 - (iv) `build.gradle`
- (c) What does APK stand for?
- (i) Android Phone Kit
 - (ii) Android Package Kit
 - (iii) Android Primary Key
 - (iv) Application Package Key
- (d) Which language is officially supported for Android development?
- (i) Python
 - (ii) Swift
 - (iii) Java
 - (iv) C++

- (e) Which language has been recommended by Google for Android development since 2019?
- (i) Kotlin
 - (ii) C#
 - (iii) JavaScript
 - (iv) Ruby
- (f) What is the purpose of `R.java` file?
- (i) To store app settings
 - (ii) To manage database
 - (iii) To link XML with Java/Kotlin code
 - (iv) To debug code
- (g) Which component is used to display a UI screen in Android?
- (i) Broadcast Receiver
 - (ii) Service
 - (iii) Activity
 - (iv) Fragment
- (h) Which method is called when an activity is first created?
- (i) `onStart()`
 - (ii) `onCreate()`
 - (iii) `onResume()`
 - (iv) `onRestart()`
- (i) Which layout arranges elements in a single column or row?
- (i) `ConstraintLayout`
 - (ii) `GridLayout`
 - (iii) `FrameLayout`
 - (iv) `LinearLayout`
- (j) Which of the following is used for local data storage in Android?
- (i) Firebase
 - (ii) MySQL
 - (iii) SQLite
 - (iv) PostgreSQL

Question 2 Define any five of the following terms:

[5 × 2 = 10]

- (a) APK
 - (b) Gradle in Android Studio
 - (c) Layout in Android Studio
 - (d) `AndroidManifest.xml`
 - (e) Activity in Android
 - (f) Intent in Android development
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Question 3 Write short answer type questions:

[2 × 5 = 10]

- (a) Describe the components of Android application development.
 - (b) Explain the use of SQLite in Android with an example.
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Question 4

[10]

Explain any five UI components of Android application development.

Question 5

[2 × 5 = 10]

- (a) What is a content provider in Android? What are the operations of the content provider?
 - (b) Explain the working of the content provider.
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Question 6 Write the SQLite query for the following statements:

[5 × 2 = 10]

- (a) Create a **student** table with attributes: ID, name, age, gender, department, and marks.
 - (b) Write an SQLite query to insert a new record with details: Name: Raj, Age: 21, Gender: male, Department: Computer Science, Marks: 85.
 - (c) Write a query to display names of the students from **student** table.
 - (d) Write a query to display all student records from table.
 - (e) Write a query to update the marks of the student with ID = 3 to 90.
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Question 7

[2 × 5 = 10]

- (a) Write a Kotlin program with a function `add` that returns the sum of two numbers.
- (b) Write a Kotlin program to print numbers from 1 to 10 using a loop statement.